

Microeconomic theory

testbank

First: true or false

1. In case of price change for a normal good, income effect must equal to price effect
2. Social surplus equals Consumer surplus minus producer surplus
3. Consumer surplus is the extra utility derived from a purchase of goods & services that has a value to the consumer greater than the market price.
4. The price ceiling must be above the equilibrium price.
5. The income effect is the fact that as the price of a good goes down, more of everything (all goods) will be demanded
6. If the two goods are complements, in case of price change of good X, the income effect is greater than the substitution effect
7. In case of inferior good, the substitution effect outweighs the income effect
8. According to cardinal utility, the more of one good consumed in a given period, the more utility generated by consuming each additional unit of the same good.
9. The budget line shows various combinations of the two goods which give equal utility to the consumer
10. When the two products are perfect substitutes, the MRS (Marginal Rate of Substitution) will decrease as one moves down the indifference curve.
11. The price Consumption curve is the curve made by joining the equilibrium quantities of two goods when the price level is constant
12. Marginal Utility is the additional satisfaction gained by consuming one more unit of a good
13. Given that the marginal utility of good X = 45 & of good Y = 65, at equilibrium the slope of budget line is equal to (-0.69)

14. The indifference curves for perfect complements will always be right angle.
15. Indifference curves never intersect
16. The law of diminishing marginal utility cannot be applied to some goods because the consumption of these goods is not continuous in nature.
17. When you are spending money on a commodity, the marginal utility of money remains constant in ordinal utility approach.
18. In cardinal utility approach, if there are more than one commodity, the condition for equilibrium of the consumer is the equality of the ratios of prices of the individual commodities to their marginal utilities (P/MU).
19. For the consumer to be in equilibrium the MRS must equal the ratio of prices of good y to good x (P_y/P_x)
20. The price Consumption curve is the curve made by joining the equilibrium quantities of two goods at different income levels.
21. According to cardinal utility, Consumer equilibrium in case of two commodities (say X and Y) is when $MU_x = MU_y$
22. According to the cardinal utility approach, the marginal utility of money decreases as the level of income increase
23. Additional utility derived from the consumption of an additional unit of a commodity is called marginal utility
24. The income consumption curve (ICC) is the locus of points of consumer equilibrium resulting when the price of good X changes
25. The consumption of inferior goods declines with the increase in income
26. When the price of the good (on the X axis) for which the demand is inelastic changes, the PCC so drawn as downward sloping.

27. In case of a normal good the substitution effect and the income effect reinforce each other
28. If the commodities are perfect substitutes, the indifference curve becomes a positively sloped straight line.
29. At each point of income consumption curve (ICC), the slope of indifference curve is equal to slope of budget line.
30. An increase in the price of good X means that the budget constraint gets steeper

Solution: true or false

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Second: CHOOSE

1. Using the following demand and supply functions in goods market:
Demand: $P = 12 - 0.25 Q_d$ Supply: $P = 5 + 0.1 Q_s$ The producer surplus =
 a. 130
 b. 7
 c. 20
 d. None of the above

2. Using the following demand and supply functions in goods market:
Demand: $P = 12 - 0.25 Q_d$ Supply: $P = 5 + 0.1 Q_s$ The total surplus =
.....
a. 200
b. 70
c. 180
d. None of the above
3. When the price of good X (the good on the X- axis) whose demand curve isdecreases, the price consumption curve is downward sloping.
a. Inelastic
b. Unitary elastic
c. Zero elastic
d. Elastic
4. If the price of good X decreases by 10% as a result quantity demanded increase by 5% therefore the PCC will be
a. downward sloping
b. upward sloping
c. backward sloping
d. horizontal
5. If the price of an inferior good decreases,
a. its quantity demanded will increase according to income effect
b. its quantity demanded will decrease according to substitution effect
c. the income effect and substitution effect work in opposite directions
d. all the above
e. none of the above
6. If the demand curve for good X (the good on the X axis) is unitary elastic, a decrease in the price of this good results in
a. no change in the quantity demanded for good X
b. an increase in the quantity demanded of the other good Y
c. a horizontal price consumption curve
d. all the above

7. In goods Market where: Demand curve: $Q_d = 420 - 2P$ Supply curve: $Q_s = 70 + 5P$ The equilibrium price =
- a. 50
 - b. 70
 - c. 140
 - d. None of the above
8. In goods Market where: Demand curve: $Q_d = 420 - 2P$ Supply curve: $Q_s = 70 + 5P$ The equilibrium quantity=
- a. 280
 - b. 320
 - c. 420
 - d. None of the above
9. Ordinal utility approach
- a. postulated that utility can be measured
 - b. assumes the constancy of the marginal Utility of money
 - c. postulates that consumer can give rank the various baskets of goods according to the satisfaction that each bundle gives him
10. A shift in budget line, is due to:
- a. change in supply
 - b. change in tastes
 - c. change in income
 - d. change in marginal utility of one of the goods
11. which of the following factors does not affect the slope & position of a consumer's budget line between 2 goods A & B
- a. shape & position of consumer's indifference curve
 - b. Consumer's income
 - c. Price of good A
 - d. price of good B

12. In case of price change of inferior good
- the income effect and substitution effect work in opposite directions
 - income effect is greater than the price effect
 - demand curve is upward sloping
 - all the above
13. When total utility reaches its maximum, marginal utility becomes
- Positive
 - Negative
 - Zero
 - All may be possible
14. Along the same indifference curve, when increasing the amount of good X (on the X axis)
- marginal utility reaches 1
 - its slope equals (P_x/P_y)
 - marginal rate of substitution of X for Y decreases
 - all the above
 - none of the above
15. According to the table, as the price of good X decrease, the price consumption curve is



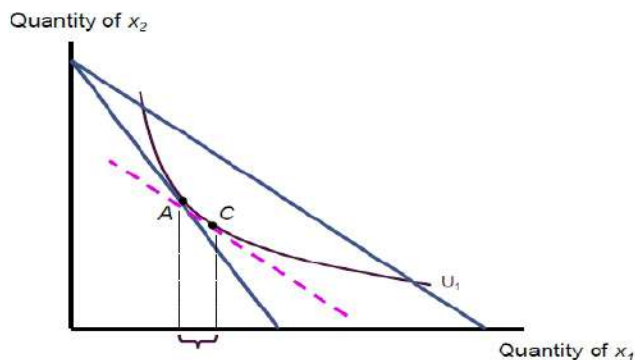
<u>P_x</u>	<u>Q_x</u>	<u>Q_y</u>
100	6	10
50	11	7
30	20	6

(P_x is the price of good x,
Q_x is the quantity of good X on the horizontal axis,
Q_y is the quantity of good Y on the vertical axis)

- upward sloping
- downward sloping
- Vertical
- horizontal

16. A change in the price of good X (on the X-axis) for which the demand is, results in upward sloping PCC

- a. inelastic
- b. elastic
- c. unitary elastic
- d. infinitely elastic



17. According to this graph, when the price of good X1 decrease, the distance between A and C represents

- a. Price effect b. income effect c. substitution effect d. utility effect

18. Which of the following is a property of indifference curve:

- a. The higher the indifference curve, the higher the level of satisfaction
- b. Two indifference curves cannot intersect each other
- c. Indifference curve is convex to origin
- d. all the above
- e. none of the above

19. When total utility is decreasing, marginal utility is

- a. Positive
- b. Zero
- c. Negative
- d. none of the above

20. If goods X & Y are normal goods & the price of good X falls
The two goods are substitutes if
- a. the substitution effect is greater than the price effect
 - b. the substitution effect is greater than the income effect
 - c. the income effect is greater than the substitution effect
 - d. none of the above
21. When the price of decreases, the PCC so drawn
is downward sloping.
- a. a giffen good
 - b. a good whose demand curve is elastic
 - c. a good whose demand curve is inelastic
 - d. a good whose demand curve is unitary elastic
22. If the price of good x falls, other things being equal, the
substitution effect
- a. can be isolated by drawing a budget line parallel to the new budget line
& tangent to the original indifference curve
 - b. can be represented graphically by movement from one point to another
along the original indifference curve
 - c. encourages the consumer to buy more of good X and less of good Y
 - d. all the above
 - e. none of the above

WORK HARD
TRUST GOD

Problems

First:

Practice Problem

Suppose that the market for gasoline can be represented by the following equations:

$$\begin{array}{ll}\text{Demand:} & P = 10 - 2Q_D \\ \text{Supply:} & P = 1 + 0.5Q_S\end{array}$$

where P is the price per gallon, and Q represents quantity of gasoline, represented in millions of gallons of gasoline consumed per year.

- Calculate the equilibrium price and quantity of gasoline.
- Concerned over high prices, the government sets a price ceiling of \$2.25 per gallon of gasoline. What is the new quantity of gasoline sold in the marketplace? Use a supply and demand diagram to illustrate your answer, showing both the original equilibrium from part (a) and the new quantity sold with the price ceiling.
- Calculate the consumer surplus and producer surplus at the initial equilibrium price and quantity from part (a).
- Calculate the new consumer surplus and producer surplus with the price ceiling of \$2.25 per gallon (part b).

Solution

- a) The equilibrium occurs where supply equals demand:

$$\begin{aligned}10 - 2Q &= 1 + 0.5Q \\ 2.5Q &= 9 \\ Q &= 9/2.5 \\ \mathbf{Q} &= \mathbf{3.6 \text{ million gallons}}\end{aligned}$$

To find the price, we substitute the equilibrium quantity into either the demand or supply equation:

Either:

$$P = 1 + 0.5(3.6) = \$2.80$$

Or:

$$P = 10 - 2(3.6) = \$2.80$$

b) The price ceiling is below the equilibrium price of gasoline. Thus, there will be excess demand for gasoline. Consumers will want to purchase more gasoline than suppliers will make available at the price of \$2.25 per gallon. Thus, we need to find out how much supply is available. We do this by plugging in \$2.25 in the supply curve and solving for Q:

$$2.25 = 1 + 0.5Q$$

$$1.25 = 0.5Q$$

$$Q = 2.5$$

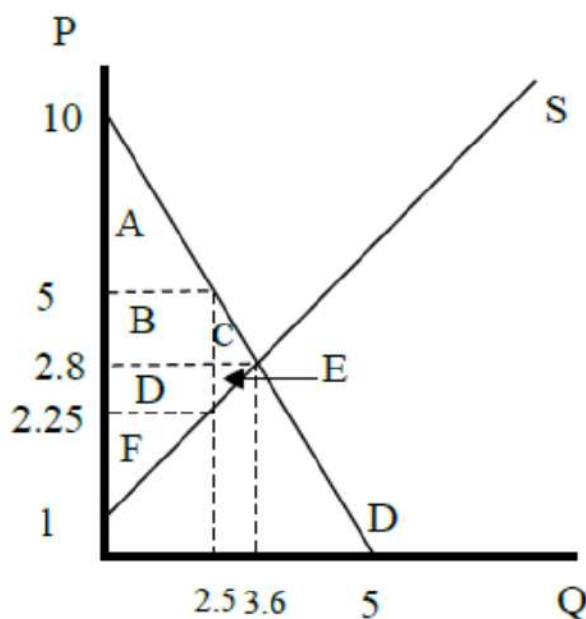
2.5 million gallons of gasoline are sold

NOTE

Because the price ceiling is lower than the equilibrium price, There will be excess demand.

To draw the graph

- The y-intercept (on the price axis) for demand is \$10, by setting $Q = 0$, (because $10 - 2(0) = 10$).
- For supply, the y-intercept is at \$1,
- Supply intersects demand at a quantity of 3.6 and a price of \$2.80.



C) At the initial equilibrium price and quantity from part (a).

Consumer surplus is the area under the demand curve and above the price. In the initial equilibrium, this includes areas A, B, & C:

This is a triangle with a height of 7.2 ($= 10 - 2.8$) and a base of 3.6.

Its area is $0.5(7.2)(3.6) = \$12.96$.

Producer surplus is the area above the supply curve and below the price producers receive. It is areas D, E, & F:

This is a triangle with a height of 1.8 ($= 2.8 - 1$) and a base of 3.6.

Its area is $0.5(1.8)(3.6) = \$3.24$.

The total consumer and producer surplus (social surplus) in part (c) is \$16.20.

d) the new consumer surplus and producer surplus with the price ceiling of \$2.25

consumer surplus is everything above the new price (\$2.25) and below the demand curve. This is areas A, B, & D (till the new quantity supplied after the price ceiling):

To find the area of A, we need to know the value at the bottom of the triangle. This is the amount that consumers are willing to pay for 2.5 million gallons of gasoline. We find this by plugging 2.5 into demand: $P = 10 - 2(2.5) = \$5$.

Thus, triangle A has a height of 5 ($= 10 - 5$) and a base of 2.5.

Its area equals $0.5(5)(2.5) = \$6.25$.

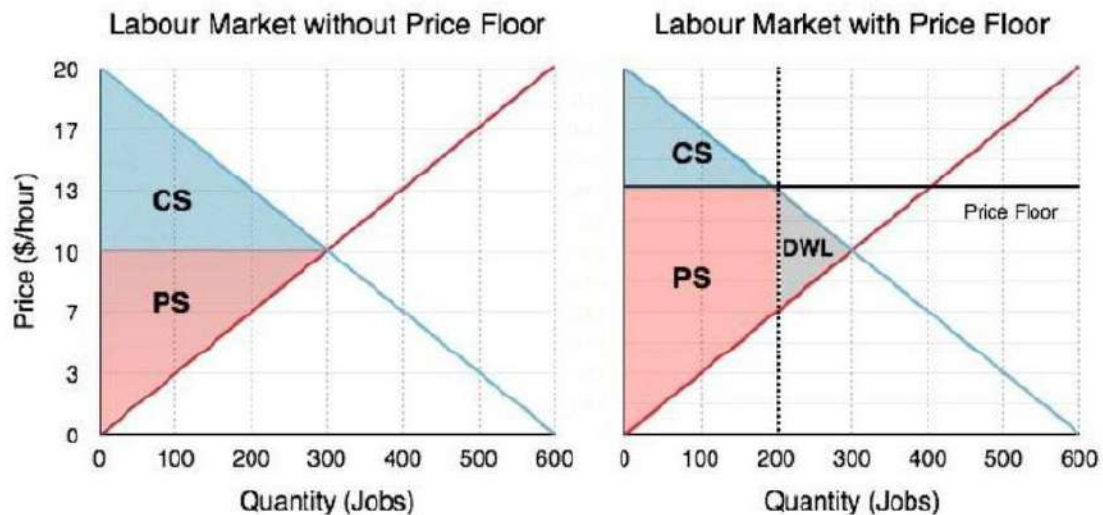
The area of the rectangle comprised of B & D has a height of 2.75 ($= 5 - 2.25$) and a length of 2.5. Its area equals $(2.75)(2.5) = \$6.875$. Thus, the total consumer surplus is **\$13.125**.

Producer surplus is just area F. This is a triangle with a height of 1.25 ($= 2.25 - 1$) and a base of 2.5. Its area is $0.5(1.25)(2.5) = \$1.5625$.

The total consumer and producer surplus (social surplus) in part (d) is \$14.6875

Problem

Based on the following graphs: Calculate consumer and producer surplus before & after price floor (minimum wage)



Before

Consumer Surplus (Blue Area): $[(20-10) \times 300]/2 = \1500

Producer Surplus (Red Area): $[(10) \times 300]/2 = \$1500$

Total Surplus: \$3000

After the price floor

Consumer Surplus (Blue Area): $[(20-13) \times 200]/2 = \700

Producer Surplus (Red Area): $[(13-7) \times 200] + (7 \times 200)/2 = \1900

Total Surplus: \$2600

Since the market surplus after the policy is less than the market surplus before, there is a deadweight loss (DWL), caused by workers who are no longer able to find jobs as firms leave the market.

$DWL = 3000 - 2600 = 400$